

6th Sem/ Electrical
Subject : Power System Protection

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Most commonly occurring fault in power system is
- a) L-G b) L-L-G
 - c) L-L-L d) L-L-L-G
- Q.2 Buchholtz relay is used for protection of
- a) Transmission line b) Alternator
 - c) Transformer d) Switch yard
- Q.3 The relay used for feeder protection is :
- a) Undervoltage relay b) Translay relay
 - c) Thermal relay d) Buchholtz relay
- Q.4 What percentage of Fault occurring in the power system in single line-to- ground fault.
- a) 50% b) 60%
 - c) 75% d) 85%

- Q.5 Ground wire is used to protect transmission lines against ____.
- a) Heating effects
 - b) Leakage current
 - c) Direct lightning strokes
 - d) All of above
- Q.6 Which of the following relay has inherent directional characteristic?
- a) Impedance type b) MHO type
 - c) Reactance type d) All of above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define Earthing.
- Q.8 The fuse element should have ____ melting point.
- Q.9 The Buchholz relay operates on the principle of ____.
- Q.10 HRC meant for ____.
- Q.11 A lightning arrestor is connected between ____ and ____.
- Q.12 The most commonly used system for the protection of alternator is ____ Protection.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Differentiate between symmetrical and unsymmetrical faults.
- Q.14 What do mean by over voltages? Explain its causes.
- Q.15 Explain the working principle of rod gap lighting arrester.
- Q.16 Differentiate between switch, isolator and circuit breaker.
- Q.17 Mention the various applications of static rely.
- Q.18 Discuss constructional features of Oil Circuit breaker.
- Q.19 What are the various types of faults that occur in underground lines?
- Q.20 Explain in brief the various methods of earthing with neat diagram.
- Q.21 What is surge absorber? How do they differ from surge diverter?
- Q.22 Discuss the differential protection of feeders.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Describe in detail the Merz Price system of protection for a 3-Phase star delta transformer.
- Q.24 Describe the construction, principle of operation and application of SF6 circuit breaker.
- Q.25 Draw a neat sketch of an induction type over current relay and describe its operation.